

Ajay S

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

Summary

Fullstack engineer with strong foundations in Data Structures & Algorithms, OOP, REST APIs, SQL, and backend development. Experienced in building API-driven applications using Java, Python, Node.js, Spring Boot, and REST APIs, with hands-on experience in database integration, API integration, debugging, Git, Docker, and AI-assisted development.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C++, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

Databases: MySQL, MongoDB, SQLite

Frontend: React.js, React Native, Bootstrap, Tailwind

Additional: Socket.IO, JWT, AWS

Backend: Node.js, Express.js, Spring Boot, Flask, FastAPI, REST APIs

Integration & APIs: REST APIs, HTTP, JSON, API Integration, Client-Server Architecture

Tools: Git, GitHub, Vercel, Docker, Postman, CI/CD

AI/ML: LangChain, FAISS, RAG, LLMs, OpenCV

Soft Skills: Flexible, Leadership, Efficient, Eager to learn

Education

Easwari Engineering College

B.E. Computer Science and Engineering — CGPA: 9.0/10

Chennai, TN

2023 – 2027

Agathiya Matric Higher Secondary School

Higher Secondary (HSC) — 95%

Chennai, TN

2022 – 2023

Professional Experience

Full-Stack Engineer Intern

July 2025 – December 2025

SRM Groups, Chennai

- Developed and maintained full-stack medical portal features using **React.js, Node.js, REST APIs, and MongoDB**, integrating frontend workflows with backend services.
- Implemented and consumed APIs for application workflows, handling **HTTP requests, JSON data, authentication, and database operations**.
- Debugged API, frontend, and backend issues and optimized workflows using **Git-based development practices**.

Projects

SemantiCache: Multi-Objective Cache Optimization | Python, FAISS, Pareto Optimization, MOSS

- Developed a multi-objective semantic cache framework using **Pareto-based admission and eviction** to optimize token savings and query volatility.
- Analyzed **261 MOSS clusters** using Spearman correlation and evaluated GPTCache, SCALM, and SemantiCache using hit ratio, token saving, staleness, and false-hit metrics.

Enterprise E-Commerce Platform | Java, Spring Boot, React.js, MySQL, Docker, AWS

- Built an enterprise-style e-commerce platform using **Java, Spring Boot REST APIs, React.js, and MySQL** for product, order, user, and payment workflows.
- Implemented **JWT authentication, RBAC, validation, layered architecture**, and automated build/deployment workflows using Docker, GitHub CI/CD, and AWS.

Real-Time Chat Application | React Native, Node.js, Socket.IO, MongoDB

- Developed a real-time messaging system using **Node.js, Socket.IO, and MongoDB** supporting concurrent multi-user and room-based communication.
- Implemented **event-driven communication, delivery acknowledgments, message synchronization**, and JWT-secured REST APIs for authentication and message management.

UPI Payment Exception Investigator | Python, FastAPI, LangChain, RAG, FAISS, LLM, SQLite

Ongoing

- Developing an AI-driven system using **Python and FastAPI** to investigate UPI payment exceptions across transaction, settlement, reversal, and PSP records.
- Built a **LangChain RAG pipeline with FAISS** for evidence retrieval and implemented deterministic workflows for validation, investigation, and financial exposure analysis.

Certifications

- **Oracle Certified Professional: Java SE 17 Developer (1Z0-829)**
- Cisco: Python Essentials, Data Science Fundamentals, AI Analytics, CCNA Levels 1–3
- **Infosys Springboard: Front-End Developer Certification**

Achievements

- **GATE 2026 Qualified (CS)**
- Solved 400+ **Data Structures & Algorithms** problems on LeetCode

Hackathons

- **Dekathon 3.0 (Finalist):** Built a platform connecting skilled, unemployed, and daily-wage workers with employers.
- **Nexathon '25 (Finalist):** Created an AI-based system for drug–drug and food–drug interaction prediction from scans.
- **Participated:** ET GenAI, HackTriX, La Trobe Hackathon, Smart India Hackathon (SIH).

Open Source Contribution

- **EthLinks:** Developed a dynamic user interface and integrated Ethereum smart contracts to streamline user interactions on the platform.
- **SMedia Influ:** Engineered a responsive full-stack web application, optimizing cross-device compatibility and user experience.